Physical Chemistry (I)

Lecture 06

Enthalpy

Thermochemistry

In the Last Class

- The first law
  - The first law of thermodynamics
  - Work
  - Heat
- Differential form of internal energy
  - Internal pressure
  - Response functions

Last class

First Law of Thermodynamics

"The internal energy of an isolated system is constant."

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$\mathrm{d}U = \delta q + \delta w$$

small change of U small q small w

Last class

Calculation of Expansion Work

$$\delta w = -p_{\mathrm{ex}}\mathrm{d}V$$

- Total work for a change from state i to state f

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\mathrm{ex}}\mathrm{d}V) = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V$$

1. Expansion against constant external pressure
2. Reversible expansion

Last class

Calculation of Heat at Const V

$$\delta q_V = C_V \mathrm{d}T$$

- Total heat for a change from state i to state f

$$q_V = \int_{\text{state } i}^{\text{state } f} \delta q_V = \int_{\text{state } i}^{\text{state } f} (C_V \mathrm{d}T) = \int_{T_i}^{T_f} C_V \, \mathrm{d}T$$

State and Path Functions

- State function: a property that depends only on the current state of a system and is independent of its history

  e.g. internal energy, pressure, volume, temperature

- Path function: a property with a value that depends on the path between two states

  e.g. heat, work

Last class

Related Mathematics

- Exact differential: an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

$$\Delta U = \int_{\text{state } i}^{\text{state } f} \mathrm{d}U$$

- Inexact differential: an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i,\ \text{path}}^{\text{state } f} \delta q$$

Use of Differentials

$$dU = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} dV + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} dT$$

$$dU = \pi_T dV + C_V dT$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Enthalpy H

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)
- Since U, p, and V are state functions, H is also a state function
- Molar enthalpy $H_{\mathrm{m}} = H/n$

Infinitesimal Change

For an infinitesimal change in the state,

$$U \rightarrow U + \mathrm{d}U$$

$$p \rightarrow p + \mathrm{d}p$$

$$V \rightarrow V + \mathrm{d}V$$

$$H \rightarrow H + \mathrm{d}H$$

From $H = U + pV$,

$$H + \mathrm{d}H = (U + \mathrm{d}U) + (p + \mathrm{d}p)(V + \mathrm{d}V)$$

Infinitesimal Change in $\boldsymbol{H}$

$$H + \mathrm{d}H = (U + \mathrm{d}U) + (p + \mathrm{d}p)(V + \mathrm{d}V)$$

$$H + \mathrm{d}H = U + \mathrm{d}U + pV + p\mathrm{d}V + V\mathrm{d}p + \cancel{\mathrm{d}p\mathrm{d}V}$$

too small

Since $H = U + pV$,

$$\mathrm{d}H = \mathrm{d}U + p\mathrm{d}V + V\mathrm{d}p$$

From the first law of thermodynamics,

$$\mathrm{d}H = \delta q + \delta w + p\mathrm{d}V + V\mathrm{d}p$$

Reversible Isobaric Process

$$\mathrm{d}H = \delta q + \delta w + p\mathrm{d}V + V\mathrm{d}p$$

For a reversible change,

$$\delta w = -p_{\mathrm{ex}}\mathrm{d}V = -p\mathrm{d}V$$

so that

$$\mathrm{d}H = \delta q - p\mathrm{d}V + p\mathrm{d}V + V\mathrm{d}p = \delta q + V\mathrm{d}p$$

At constant pressure ($\mathrm{d}p = 0$),

$$\mathrm{d}H = \delta q_p$$

Enthalpy Change

$$\mathrm{d}H = \delta q_p$$

Integration leads to

$$\Delta H = \int_{\text{state } i}^{\text{state } f} \mathrm{d}H = \int_{\text{state } i}^{\text{state } f} \delta q_p = q_p$$

Heat Capacity at Constant p

Since $\delta q_p = C_p \mathrm{d}T$,

$$\mathrm{d}H = \delta q_p = C_p \mathrm{d}T$$

which means

$$C_p = \left(\frac{\partial H}{\partial T}\right)_p$$

The change in enthalpy is

$$\Delta H = \int_{\text{state } i}^{\text{state } f} \mathrm{d}H = \int_{\text{state } i}^{\text{state } f} \left(C_p \mathrm{d}T\right) = \int_{T_i}^{T_f} C_p \, \mathrm{d}T$$

At Constant p

$$\Delta H = \int_{T_i}^{T_f} C_p \, \mathrm{d}T$$

If C_p is constant,

$$\Delta H = \int_{T_i}^{T_f} C_p \, \mathrm{d}T = C_p \int_{T_i}^{T_f} \mathrm{d}T = C_p\left(T_f - T_i\right) = C_p \Delta T$$

Since $\Delta H = q_p$ at constant pressure,

$$q_p = C_p \Delta T$$

Changes in Enthalpy

For $H = H(p, T)$,

$$\mathrm{d}H = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{?} \mathrm{d}p + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} \mathrm{d}T$$

Let $\mathrm{d}H = 0$ (constant enthalpy).

$$\left(\frac{\partial H}{\partial p}\right)_T \mathrm{d}p = -C_p \mathrm{d}T$$

Constant Enthalpy

At constant enthalpy,

$$\left(\frac{\partial H}{\partial p}\right)_T \mathrm{d}p = -C_p \mathrm{d}T$$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \left(\frac{\partial T}{\partial p}\right)_H$$

Define the Joule-Thomson coefficient: $\mu = (\partial T/\partial p)_H$

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p \mu$$

Changes in Enthalpy

For $H = H(p, T)$,

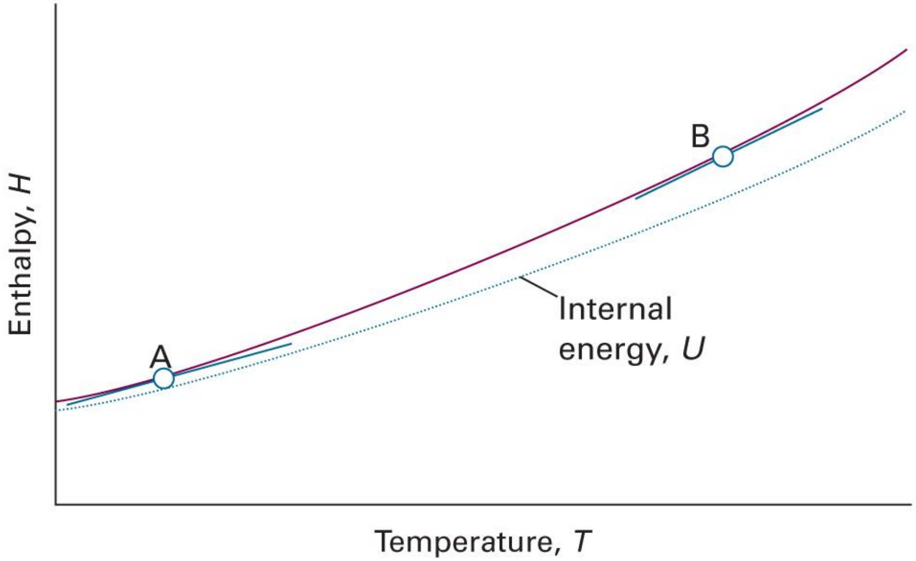
$$\mathrm{d}H = \underbrace{\left(\frac{\partial H}{\partial p}\right)_T}_{-C_p\mu} \mathrm{d}p + \underbrace{\left(\frac{\partial H}{\partial T}\right)_p}_{C_p} \mathrm{d}T$$

$$\mathrm{d}H = -C_p\mu\mathrm{d}p + C_p\mathrm{d}T$$

(Compare: $\mathrm{d}U = \pi_T\mathrm{d}V + C_V\mathrm{d}T$)

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_p$$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 2B.3)

Case of a Perfect Gas

$$C_p - C_V = \left(\frac{\partial H}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_p$$

For a perfect gas,

$$H = U + pV = U + nRT$$

so that

$$C_p - C_V = \left(\frac{\partial (U + nRT)}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_p$$

$$= \left(\frac{\partial U}{\partial T}\right)_p + nR - \left(\frac{\partial U}{\partial T}\right)_p = nR$$

Adiabatic Changes

$$\delta q = 0$$

$$\mathrm{d}U = \delta q + \delta w = \delta w$$

"Adiabatic expansion" $\longrightarrow U$ drop $\longrightarrow T$ drop

Work for Perfect Gas

$$\mathrm{d}U = \delta w$$

For a perfect gas, $\mathrm{d}U = C_V \mathrm{d}T$

$$C_V \mathrm{d}T = \delta w$$

Integration leads to

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} C_V \, \mathrm{d}T = \int_{T_i}^{T_f} C_V \, \mathrm{d}T$$

If C_V is constant,

$$w = C_V\left(T_f - T_i\right)$$

Perfect Gas, Reversible Change

$$\mathrm{d}U = \delta w$$

- Perfect gas: $\mathrm{d}U = C_V \mathrm{d}T$
- Reversible change: $\delta w = -p\mathrm{d}V$

$$C_V \mathrm{d}T = -p\mathrm{d}V$$

Since $p = nRT/V$,

$$\frac{C_V}{T}\mathrm{d}T = -\frac{nR}{V}\mathrm{d}V$$

Integration

$$\int_{\text{state } i}^{\text{state } f} \frac{C_V}{T}\,\mathrm{d}T = -\int_{\text{state } i}^{\text{state } f} \frac{nR}{V}\,\mathrm{d}V$$

$$\int_{T_i}^{T_f} \frac{C_V}{T}\,\mathrm{d}T = -\int_{V_i}^{V_f} \frac{nR}{V}\,\mathrm{d}V$$

If C_V is constant,

$$C_V \int_{T_i}^{T_f} \frac{1}{T}\,\mathrm{d}T = -nR \int_{V_i}^{V_f} \frac{1}{V}\,\mathrm{d}V$$

$$C_V\left(\ln T_f - \ln T_i\right) = -nR\left(\ln V_f - \ln V_i\right)$$

Simplification

$$C_V\left(\ln T_f - \ln T_i\right) = -nR\left(\ln V_f - \ln V_i\right)$$

$$\frac{C_V}{nR}\ln\frac{T_f}{T_i} = -\ln\frac{V_f}{V_i} = \ln\frac{V_i}{V_f}$$

Let $c = C_V/nR$;

$$\frac{C_V}{nR}\ln\frac{T_f}{T_i} = c\ln\frac{T_f}{T_i} = \ln\left(\frac{T_f}{T_i}\right)^c$$

$$\left(\frac{T_f}{T_i}\right)^c = \frac{V_i}{V_f}$$

Volume-Temperature

For $c = C_V/nR$,

$$\left(\frac{T_f}{T_i}\right)^c = \frac{V_i}{V_f}$$

Therefore,

$$V_i T_i^c = V_f T_f^c$$

or,

$$VT^c = \text{constant}$$

From $\boldsymbol{T}$ to $\boldsymbol{p}$

$$V_i T_i^c = V_f T_f^c$$

For a perfect gas, $T = pV/nR$

$$V_i \left(\frac{p_i V_i}{nR}\right)^c = V_f \left(\frac{p_f V_f}{nR}\right)^c$$

$$p_i^c V_i^{c+1} = p_f^c V_f^{c+1}$$

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

Volume-Pressure

For $c = C_V/nR$,

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

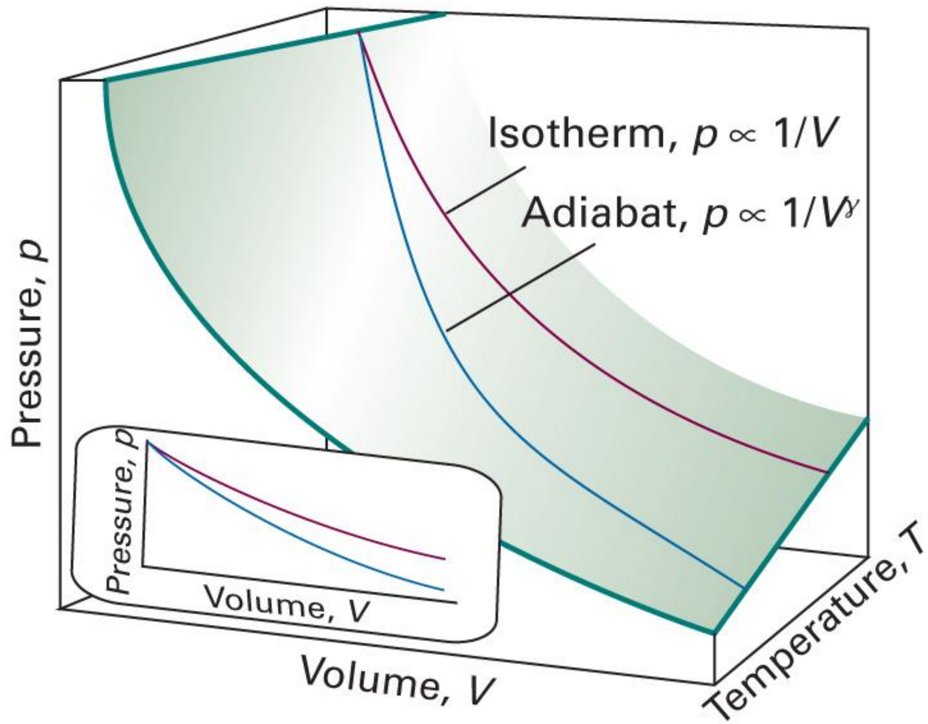
Let $\gamma = (c+1)/c = 1 + 1/c$;

$$p_i V_i^{\gamma} = p_f V_f^{\gamma}$$

or,

$$pV^{\gamma} = \text{constant}$$

Isotherm and Adiabat



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 2E.2)

Monatomic Perfect Gas

$$VT^c = \text{constant}$$

$$pV^\gamma = \text{constant}$$

$$c = \frac{C_V}{nR} = \frac{(3/2)nR}{nR} = \frac{3}{2}$$

$$\gamma = 1 + \frac{1}{c} = 1 + \frac{2}{3} = \frac{5}{3}$$

Summary for Chapter 3 (2)

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Enthalpy

$$H = U + pV$$

$$\mathrm{d}H = \delta q_p$$

Differential form of U

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T$$

Differential form of H

$$\mathrm{d}H = \left(\frac{\partial H}{\partial p}\right)_T \mathrm{d}p + \left(\frac{\partial H}{\partial T}\right)_p \mathrm{d}T$$

$$C_V = (\partial U/\partial T)_V$$

C_V and C_p

- Perfect gas
- Adiabatic change: $\delta q = 0$

$$C_p = (\partial H/\partial T)_p$$

Thermochemistry

"The study of the energy transferred as heat during the course of chemical reactions"

$$\underbrace{\mathrm{A} + \mathrm{B}}_{\text{state } i} \longrightarrow \underbrace{\mathrm{C} + \mathrm{D}}_{\text{state } f}$$

$$q_V = \Delta U$$

$$q_p = \Delta H$$

Endothermic and Exothermic

A + B ⟶ C + D, $\Delta H = q_p$

$\Delta H > 0$: endothermic reaction

$\Delta H < 0$: exothermic reaction

Standard Enthalpy, $\boldsymbol{H}^{\ominus}$

Enthalpy H of a substance in its standard state

- Standard state of a substance: its pure form at 1 bar
  - Defined for a specific temperature
  - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^{\ominus} = H_f^{\ominus} - H_i^{\ominus}$

Example: Vaporization of Water

$$H_2O(l) \rightarrow H_2O(g)$$

$$\Delta H^{\ominus}(373\ \text{K}) = 40.66\ \text{kJ mol}^{-1}$$

pure liquid water → pure gas water

at 1 bar and 373 K

Not in Atkins

H and ΔH

$$\Delta H = H(\text{product}) - H(\text{reactant})$$

$$H = U + pV$$

One can determine absolute H for a perfect gas

But what about real substances?

(liquids, solids, solutions, ...)

Instead, we focus on the "change," ΔH, in chemistry

For Typical Changes

- The standard enthalpy change is called the standard enthalpy of [transition]
- Physical change
  - Vaporization $\Delta_{\mathrm{vap}}H^{\ominus}$: standard enthalpy of vaporization
  - Fusion $\Delta_{\mathrm{fus}}H^{\ominus}$: standard enthalpy of fusion
- Chemical change
  - Reaction $\Delta_{\mathrm{r}}H^{\ominus}$: standard enthalpy of reaction
  - Combustion $\Delta_{\mathrm{c}}H^{\ominus}$: standard enthalpy of combustion

Several Types of Enthalpies

Transition	Process	Symbol*
Transition	Phase $\alpha \rightarrow$ phase β	$\Delta_{trs}H$
Fusion	$s \rightarrow l$	$\Delta_{fus}H$
Vaporization	$l \rightarrow g$	$\Delta_{vap}H$
Sublimation	$s \rightarrow g$	$\Delta_{sub}H$
Mixing	Pure $\rightarrow$ mixture	$\Delta_{mix}H$
Solution	Solute $\rightarrow$ solution	$\Delta_{sol}H$
Hydration	$X^{\pm}(g) \rightarrow X^{\pm}(aq)$	$\Delta_{hyd}H$
Atomization	Species(s, l, g) $\rightarrow$ atoms(g)	$\Delta_{at}H$
Ionization	$X(g) \rightarrow X^{+}(g) + e^{-}(g)$	$\Delta_{ion}H$
Electron gain	$X(g) + e^{-}(g) \rightarrow X^{-}(g)$	$\Delta_{eg}H$
Reaction	Reactants $\rightarrow$ products	$\Delta_{r}H$
Combustion	Compound(s, l, g) + $O_2(g) \rightarrow CO_2(g) + H_2O(l, g)$	$\Delta_{c}H$
Formation	Elements $\rightarrow$ compound	$\Delta_{f}H$
Activation	Reactants $\rightarrow$ activated complex	$\Delta^{\ddagger}H$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Table 2C.2)

Stoichiometry: Mole Method

$$\nu_{\mathrm{A}}\mathrm{A} + \nu_{\mathrm{B}}\mathrm{B} \longrightarrow \nu_{\mathrm{C}}\mathrm{C} + \nu_{\mathrm{D}}\mathrm{D}$$

means

"ν_{A} moles of A and ν_{B} moles of B react

to produce ν_{C} moles of C and ν_{D} moles of D."

(ν_i: stoichiometric coefficients)

Standard Reaction Enthalpy

$$\nu_{\mathrm{A}}\mathrm{A} + \nu_{\mathrm{B}}\mathrm{B} \longrightarrow \nu_{\mathrm{C}}\mathrm{C} + \nu_{\mathrm{D}}\mathrm{D}$$

- Standard reaction enthalpy $\Delta_{\mathrm{r}}H^{\ominus}$: standard enthalpy change $\Delta H^{\ominus}$ for the reaction per …
  - ν_{A} moles of consumed A
  - ν_{B} moles of consumed B
  - ν_{C} moles of produced C
  - ν_{D} moles of produced D

Example

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

$$\Delta_r H^\ominus = -890 \text{ kJ mol}^{-1}$$

The enthalpy change is -890 kJ at 1 bar and 298.15 K per

- 1 mol of $CH_4(g)$ in its pure form
- 2 mol of $O_2(g)$ in its pure form
- 1 mol of $CO_2(g)$ in its pure form
- 2 mol of $H_2O(l)$ in its pure form

Formal Definition

$$\Delta_{\mathrm{r}}H^{\ominus} = \sum_{\text{products } i} \nu(i)H_{\mathrm{m}}^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)H_{\mathrm{m}}^{\ominus}(j)$$

e.g. $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

$$\Delta_{\mathrm{r}}H^{\ominus} = \nu(\mathrm{CO_2})H_{\mathrm{m}}^{\ominus}(\mathrm{CO_2}, g) + \nu(\mathrm{H_2O})H_{\mathrm{m}}^{\ominus}(\mathrm{H_2O}, l)$$

$$-\nu(\mathrm{CH_4})H_{\mathrm{m}}^{\ominus}(\mathrm{CH_4}, g) - \nu(\mathrm{O_2})H_{\mathrm{m}}^{\ominus}(\mathrm{O_2}, g)$$

$$= H_{\mathrm{m}}^{\ominus}(\mathrm{CO_2}, g) + 2H_{\mathrm{m}}^{\ominus}(\mathrm{H_2O}, l) - H_{\mathrm{m}}^{\ominus}(\mathrm{CH_4}, g) - 2H_{\mathrm{m}}^{\ominus}(\mathrm{O_2}, g)$$

Enthalpy and Temperature

$$\mathrm{d}H = C_p \mathrm{d}T$$

$$\int_{\text{state } i}^{\text{state } f} \mathrm{d}H = \int_{\text{state } i}^{\text{state } f} C_p \, \mathrm{d}T$$

$$H_f - H_i = \int_{T_i}^{T_f} C_p \, \mathrm{d}T$$

$$H(T_f) - H(T_i) = \int_{T_i}^{T_f} C_p \, \mathrm{d}T$$

Temperature Dependence

For $\nu_{\mathrm{A}}\mathrm{A} + \nu_{\mathrm{B}}\mathrm{B} \longrightarrow \nu_{\mathrm{C}}\mathrm{C} + \nu_{\mathrm{D}}\mathrm{D}$,

$$\Delta_{\mathrm{r}}H^{\ominus}(T_f) = \nu_{\mathrm{C}}H_{\mathrm{m}}^{\ominus}(\mathrm{C},T_f) + \nu_{\mathrm{D}}H_{\mathrm{m}}^{\ominus}(\mathrm{D},T_f) - \nu_{\mathrm{A}}H_{\mathrm{m}}^{\ominus}(\mathrm{A},T_f) - \nu_{\mathrm{B}}H_{\mathrm{m}}^{\ominus}(\mathrm{B},T_f)$$

Since $H(T_f) = H(T_i) + \int_{T_i}^{T_f} C_p\,\mathrm{d}T$,

$$H_{\mathrm{m}}^{\ominus}(T_f) = H_{\mathrm{m}}^{\ominus}(T_i) + \int_{T_i}^{T_f} C_{p,\mathrm{m}}^{\ominus}\,\mathrm{d}T$$

$$\Delta_{\mathrm{r}}H^{\ominus}(T_f)$$

$$= \nu_{\mathrm{C}}\left\{H_{\mathrm{m}}^{\ominus}(\mathrm{C},T_i) + \int_{T_i}^{T_f} C_{p,\mathrm{m}}^{\ominus}(\mathrm{C})\,\mathrm{d}T\right\} + \nu_{\mathrm{D}}\left\{H_{\mathrm{m}}^{\ominus}(\mathrm{D},T_i) + \int_{T_i}^{T_f} C_{p,\mathrm{m}}^{\ominus}(\mathrm{D})\,\mathrm{d}T\right\}$$

$$- \nu_{\mathrm{A}}\left\{H_{\mathrm{m}}^{\ominus}(\mathrm{A},T_i) + \int_{T_i}^{T_f} C_{p,\mathrm{m}}^{\ominus}(\mathrm{A})\,\mathrm{d}T\right\} - \nu_{\mathrm{B}}\left\{H_{\mathrm{m}}^{\ominus}(\mathrm{B},T_i) + \int_{T_i}^{T_f} C_{p,\mathrm{m}}^{\ominus}(\mathrm{B})\,\mathrm{d}T\right\}$$

Temperature Dependence

$$\Delta_r H^\ominus(T_f) = \nu_C H_m^\ominus(C, T_i) + \nu_D H_m^\ominus(D, T_i) - \nu_A H_m^\ominus(A, T_i) - \nu_B H_m^\ominus(B, T_i)$$

$$+ \int_{T_i}^{T_f} \underbrace{\left\{\nu_C C_{p,m}^\ominus(C) + \nu_D C_{p,m}^\ominus(D) - \nu_A C_{p,m}^\ominus(A) - \nu_B C_{p,m}^\ominus(B)\right\}}_{\Delta_r C_p^\ominus} dT$$

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus \, dT$$

Kirchhoff's Law

$$\Delta_\mathrm{r} H^{\ominus}(T_f) = \Delta_\mathrm{r} H^{\ominus}(T_i) + \int_{T_i}^{T_f} \Delta_\mathrm{r} C_p^{\ominus}\,\mathrm{d}T$$

where

$$\Delta_\mathrm{r} C_p^{\ominus} = \sum_{\text{products } i} \nu(i) C_{p,\mathrm{m}}^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j) C_{p,\mathrm{m}}^{\ominus}(j)$$

If $\Delta_\mathrm{r} C_p^{\ominus}$ is constant,

$$\Delta_\mathrm{r} H^{\ominus}(T_f) = \Delta_\mathrm{r} H^{\ominus}(T_i) + \Delta_\mathrm{r} C_p^{\ominus} \times (T_f - T_i)$$

Problem in $\Delta_r H^\ominus$

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) H_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) H_m^\ominus(j)$$

Can we determine absolute $H_m^\ominus(i)$?

Hess's Law

"The standard reaction enthalpy is

the sum of the values for the individual reactions

into which the overall reaction may be divided."

This holds because H is a state function

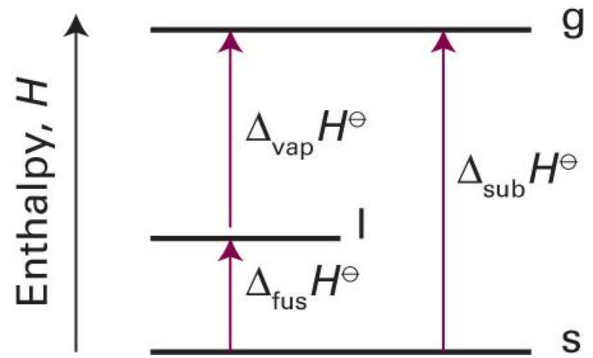
H as State Function

ΔH is always same if the initial and final states are same

Path 1: $H_2O(s) \longrightarrow H_2O(g)$, $\Delta_{\mathrm{sub}}H^{\ominus}$

Path 2: (1) $H_2O(s) \longrightarrow H_2O(l)$, $\Delta_{\mathrm{fus}}H^{\ominus}$

(2) $H_2O(l) \longrightarrow H_2O(g)$, $\Delta_{\mathrm{vap}}H^{\ominus}$

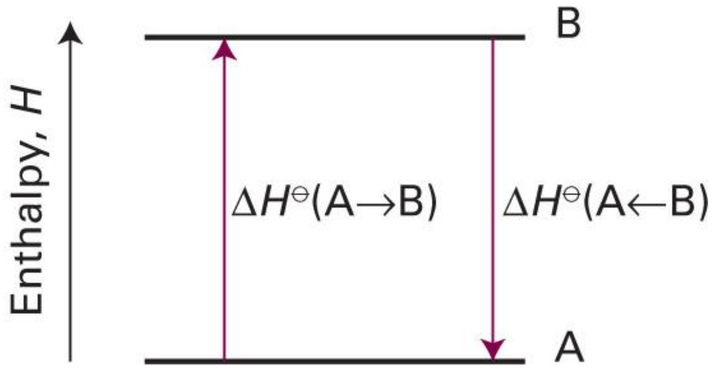


$$\Delta_{\mathrm{sub}}H^{\ominus} = \Delta_{\mathrm{fus}}H^{\ominus} + \Delta_{\mathrm{vap}}H^{\ominus}$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., p. 51)

H as State Function

$$\Delta H^{\ominus}(\text{forward}) = -\Delta H^{\ominus}(\text{reverse})$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., p. 51)

Topic 2C

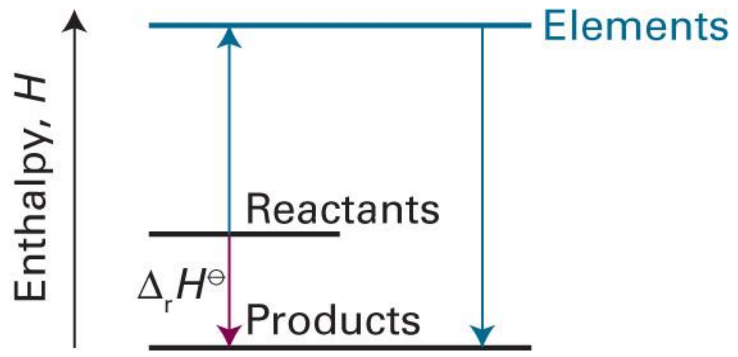
Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_f H^\ominus$ of a substance: $\Delta_r H^\ominus$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

e.g. $\Delta_f H^\ominus$ of liquid benzene at 298 K: +49.0 kJ mol^{-1}

$$6C(s, \text{graphite}) + 3H_2(g) \longrightarrow C_6H_6(l)$$

$\Delta_\mathrm{f}\boldsymbol{H}^{\boldsymbol{\ominus}}$ and $\Delta_\mathrm{r}\boldsymbol{H}^{\boldsymbol{\ominus}}$



$$\Delta_\mathrm{r}H^\ominus = \sum_{\text{products } i} \nu(i)\Delta_\mathrm{f}H^\ominus(i) - \sum_{\text{reactants } j} \nu(j)\Delta_\mathrm{f}H^\ominus(j)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., p. 53)

$\Delta_f \boldsymbol{H}^{\boldsymbol{\Theta}}$ of Elements

- For an element …
  - in its reference state, $\Delta_f H^{\Theta} = 0$ (for all T)

  C(*s*, graphite) ⟶ C(*s*, graphite), $\Delta_f H^{\Theta}$ = 0

  - in the other state, $\Delta_f H^{\Theta} > 0$

  C(*s*, graphite) ⟶ C(*s*, diamond), $\Delta_f H^{\Theta}$(298 K) = 1.895 kJ mol^{-1}

- Reference state of phosphorus (P): white phosphorus
  - not the most stable, but the most reproducible

  P(*s*, white) ⟶ P(*s*, red), $\Delta_f H^{\Theta}$(298 K) = -17.46 kJ mol^{-1}

$\Delta_f H^\ominus$ of Ions

- $\Delta_f H^\ominus$ of ions: *e.g.* Na(s) ⟶ Na^+(*aq*) + e^-?

Impossible to isolate Na^+(*aq*)

- Define $\Delta_f H^\ominus$ for one ion, and use it to determine others

$$\Delta_f H^\ominus(H^+, aq) = 0$$

e.g. HBr(*aq*) ⟶ H^+(*aq*) + Br^-(*aq*)

$$\underbrace{\Delta_f H^\ominus(\mathrm{HBr}, aq)}_{\text{measurable}} = \underbrace{\Delta_f H^\ominus(\mathrm{H}^+, aq)}_{=0} + \Delta_f H^\ominus(\mathrm{Br}^-, aq)$$

Some Values of $\Delta_f H^\ominus$

	$\Delta_f H^\ominus/(\text{kJ mol}^{-1})$
$H_2O(l)$	−285.83
$H_2O(g)$	−241.82
$NH_3(g)$	−46.11
$N_2H_4(l)$	+50.63
$NO_2(g)$	+33.18
$N_2O_4(g)$	+9.16
$NaCl(s)$	−411.15
$KCl(s)$	−436.75

	$\Delta_f H^\ominus/(\text{kJ mol}^{-1})$
$CH_4(g)$	−74.81
$C_6H_6(l)$	+49.0
$C_6H_{12}(l)$	−156
$CH_3OH(l)$	−238.66
$CH_3CH_2OH(l)$	−277.69

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Tables 2C.4 and 2C.5)

Summary for Chapter 3 (3)

Enthalpy

$H = U + pV$

Standard enthalpy $H^{\ominus}$

Standard enthalpy change $\Delta H^{\ominus}$

Heat capacity

$C_p = (\partial H/\partial T)_p$

Hess's law

Kirchhoff's law

Standard reaction enthalpy $\Delta_r H^{\ominus}$

$$\Delta_r H^{\ominus} = \sum_{\text{product } i} \nu(i) H_m^{\ominus}(i) - \sum_{\text{reactant } j} \nu(j) H_m^{\ominus}(j)$$

Standard enthalpy of formation $\Delta_f H^{\ominus}$

$$\Delta_r H^{\ominus} = \sum_{\text{product } i} \nu(i) \Delta_f H^{\ominus}(i) - \sum_{\text{reactant } j} \nu(j) \Delta_f H^{\ominus}(j)$$